Data Efficiency Assessment of Generative Adversarial Networks for Critical Heat Flux Synthetic Data Generation

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ABSTRACT

This study investigates the application of generative artificial intelligence techniques, particularly conditional generative adversarial networks (cGAN), in real-world engineering contexts, with a specific focus on synthetic data generation for critical heat flux (CHF). Utilizing a dataset comprising more than 20,000 real experimental CHF measurements, we conduct a series of experiments to examine cGAN's behavior. These experiments encompass varying sizes of the training dataset, training cGAN on data from diverse experimental sources to generate new data on unseen experimental setups, and assessing the impact of excluding various input features on cGAN's data generation accuracy. Our findings underscore the pronounced data dependency of cGAN for reliable performance, with decreased efficacy observed with smaller training dataset sizes. Notably, cGAN exhibits varying performance when trained on data from different experiments, with superior predictive capabilities observed for certain experiment sources compared to others. For instance, when cGAN was trained on data from Smolin et al.'s experiments or Zenkevich et al., it exhibited relatively good performance in generating the data from Becker et al., Kirillov et al., and Alekseev et al. experiments. In contrast, when trained with Alekseev et al.'s data and tasked with generating other experimental setups, cGAN showed notably poor performance. In both scenarios, cGAN's performance was inferior compared to training on samples from all experiments concurrently. A feature importance analysis highlights the significant influence of parameters such as mass flux and heated length on accurate CHF generation, while other parameters like diameter and pressure have less impact. Inlet temperature is identified as a moderating factor by cGAN.

KEYWORDS
Generative AI, Generative Adversarial Networks, Critical Heat Flux, Data Augmentation

1. INTRODUCTION

Artificial intelligence and machine learning (AI/ML) is currently at the forefront in reshaping traditional practices in almost every sector, from healthcare to finance and education to manufacturing by unlocking new possibilities and promising efficiency, innovation, and unparalleled advancements [1]. It has a far-reaching impact across various domains of the engineering industry, including intelligent design and simulation, autonomous systems and predictive maintenance, data augmentation, documentation, quality control and inspection processes, environmental impact assessment, and fostering sustainable engineering practices. However, in tandem with the myriad advantages, there are a multitude of challenges. One prominent challenge is data quality and availability. AI/ML algorithms heavily depend on large datasets for training and decision-making [2]. Ensuring the quality, relevance, and availability of diverse datasets poses a significant challenge, especially in domains where data may be scarce or sensitive like in engineering domains.

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One potential solution that stands out most for the significant challenge posed by a shortage of training datasets is generative AI (GenAI). The GenAI domain is dedicated to creating algorithms and models that can produce synthetic data that closely mimic real-world data [3]. It has vastly influenced many industries, including media, marketing, education, medicine, architecture, software development, and the nuclear industry [4]. It can be used to generate meaningful new content, which has paved the way for diverse applications, such as creating images and videos, generating text, augmenting data, composing poems and music, and developing chatbots with human-like characteristics.

The popular variants of GenAI models include generative adversarial networks (GANs), variational autoencoders, normalizing flow, and transformers. In 2013, Diederik P. Kingma and Max Welling introduced variational autoencoders, a special autoencoder based on stochastic variational inference and deep learning [5]. It is comprised of three parts: encoder, latent space, and decoder. Encoder compresses the input data into a lower dimensional latent space. The decoder then decompresses and tries to recreate the input data from the latent space. GAN was introduced by Goodfellow in 2014, which is a novel generative model based on a minimax game theory [6]. It has two networks: generator and discriminator. The generator creates as realistic data as possible to deceive the discriminator, which then attempts to distinguish fake samples from real ones. The training process continues until the generator succeeds in fooling the discriminator to no longer distinguish the fake data [7]. Although the framework of normalizing flow was introduced before, it was primarily popularized by Rezende and Mohamed in their 2015 article "Variational Inference with Normalizing Flows" [8]. Normalizing flows employ a series of invertible transformations to map a simple distribution to a more complex distribution, enabling the generation of realistic data samples [9]. The transformer architecture was introduced by Vaswani et al. [10]. Transformer architecture is based on the self-attention mechanism, which allows the model to weigh different parts of the input sequence differently when making predictions. We will focus this paper on GAN as a promising variant of GenAI.

Due to the wide range applications, various types of GANs have been proposed with a specific purpose [11], including conditional GAN (cGAN), deep convolutional GAN, Laplacian GAN, progressive-growing GAN, cycle GAN, and Wasserstein GAN [12]. For example, Wang and his research team used cGAN to generate artificial fault samples for a mechanical fault diagnosis in bearings and gearboxes [13]. Another variation, cycle Gan, was deployed by Branikas et al. to generate realistic images of cracks for visual defect detection in Advanced gas-cooled reactor cores [14].

To fully exploit the potential of deep learning and GenAI, a large training dataset is imperative. It has already been established that the larger the datasets are, the better results that can be obtained from deep learning [15], which is not always the case for most engineering applications [16]. This situation is even worse for the nuclear industry, where the experiments are costly and simulations are computationally expensive [17, 18]. To overcome this hurdle, researchers are seeking data augmentation to enlarge the real dataset and synthesize as realistic training data as possible. However, limited research has been conducted to determine the data needed for GenAI techniques to create new samples without losing accuracy. In this paper, we implement cGAN to augment the critical heat flux (CHF) data published by the U.S. Nuclear Regulatory Commission (NRC). The database contains 21,453 CHF measurements in vertical water-cooled uniformly heated tubes collected from 59 different sources over a span of 60 years [19]. Our objective is to generate synthetic CHF data that is as realistic as possible using cGAN while assessing its data needs, i.e., *how much data and information cGAN requires from real experiments to generate high-quality synthetic CHF data*.

The structure of the remaining sections of this paper is organized as follows here. Section 2 describes the training data collection process. The deep generative model implemented in this paper is presented in Section 3. Section 4 provides the results of data augmentation by cGAN and the performance assessment

of cGAN data needs based on the NRC CHF measurements. Finally, the conclusions drawn from this study and the prospective approach of future research are highlighted in Section 5.

2. DATA COLLECTION AND PROCESSING

This section describes the CHF dataset used in the analysis. For a nuclear reactor, one of the major safety limits is the local heat flux. Although higher heat flux is desirable for higher enrichment, it cannot be increased indefinitely. The limit is referred to as CHF. After this value, the heat transfer coefficient deteriorates significantly. This leads to higher reactor temperature and fuel failure. Depending on the operating condition of the nuclear reactor, this phenomenon is also known as boiling crisis, departure from nucleate boiling, and dryout. Over the years, CHF for vertical water-cooled tubes has been measured in many experiments worldwide. These experiments are compiled to create CHF look-up tables that are used widely for predicting the CHF. In 2019, the NRC published a database used to develop the widely known 2006 Groeneveld CHF lookup table [19]. This dataset comprises experimental data gathered over a 60 year period, based on a variety of CHF determination techniques, including visual identification, physical burnout, changes in test section resistances, and thermocouple usage. A cleaned-up version of the NRC database was provided to the Organisation for Economic Co-operation and Development (OECD) Nuclear Energy Agency for benchmarking purposes [20]. This dataset includes 21,453 CHF measurement points. All references and an overview of all test facilities that have contributed to developing the NRC CHF database are documented in [19]. The available data consists of: boundary conditions: Pressure (P), mass flux (G), inlet temperature (T_{in}), geometrical parameters: Test section diameter (D) and heated length (L), and parameters derived from measurements and water properties: Outlet equilibrium quality (X), inlet enthalpy (H_{in}), and critical heat flux (CHF).

This study uses five parameters, including boundary conditions and geometrical parameters, as conditional GAN input parameters, while CHF is the parameter to be generated by GAN. The parameter ranges of the dataset used in this study are demonstrated in Table I.

Table I. Parameter spans of the database used in this study

Variable	Diameter (mm)	Length (m)	Pressure (kPa)	Mass Flux (kg/m^{2}s)	Temperature (°C)	CHF (kW/m^2)
Minimum	2.39	0.07	100.0	17.7	9.0	130.0
Maximum	16.0	15.0	20,000.0	7,712.0	353.62	13,345.0

3. METHODOLOGY

GANs are a class of ML models that consist of two neural networks: the generator $\mathcal{G}$ and the discriminator $\mathcal{D}$. They have gained significant attention due to their ability to generate synthetic data that closely resembles real data distributions. The generator takes as input a random noise vector z sampled from a prior distribution $p_z(z)$ and produces synthetic data samples $\hat{x} = \mathcal{G}(z)$. The discriminator, on the other hand, aims to distinguish between real data samples x from the true data distribution $p_{\text{data}}(x)$ and fake samples $\hat{x}$ produced by the generator. The objective of the generator is to maximize the probability that the discriminator misclassifies its generated samples as real:

$$\min_{\mathcal{G}} \max_{\mathcal{D}} V(\mathcal{D}, \mathcal{G}) = \mathbb{E}_{x \sim p_{\text{data}}(x)}[\log \mathcal{D}(x)] + \mathbb{E}_{z \sim p_z(z)}[\log(1 - \mathcal{D}(\mathcal{G}(z)))] \tag{1}$$

where $V(\mathcal{D}, \mathcal{G})$ represents the value function or the adversarial loss. Here, the first term in the objective function represents the log probability that the discriminator correctly classifies real data as real, and the second term represents the log probability that the discriminator incorrectly classifies fake data as real. Unlike vanilla GAN, cGANs are not completely unsupervised in their training methods. Instead, cGAN architectures implement class labels or labeled data to execute specific tasks effectively. For creating a cGAN structure, a small adjustment is applied to the vanilla GAN architecture by adding a "y-label" to both the discriminator and generator networks. This adjustment transforms the previous probabilities into conditional probabilities. With this modification, the training process guarantees that the generator produces outputs aligned with the specified labels, which are provided as conditions. Similarly, the discriminator scrutinizes the authenticity of the generated output, ensuring it matches the expected label. Consequently, after training, providing a specific input label will yield the desired output from the generative network. Figure 1 illustrates the typical structure of cGANs.

$$\min_{\mathcal{G}} \max_{\mathcal{D}} V(\mathcal{D}, \mathcal{G}) = \mathbb{E}_{x \sim p_{\text{data}}(x)}[\log \mathcal{D}(x|y)] + \mathbb{E}_{z \sim p_z(z)}[\log(1 - \mathcal{D}(\mathcal{G}(z|y)))] \tag{2}$$

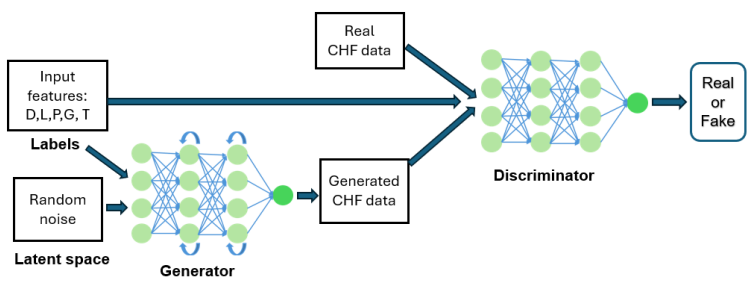

Figure 1. A typical structure of cGAN

To assess the accuracy of the generated synthetic data, cGAN is implemented in this study. In this case, all the input parameters are fixed (used as conditions) and only the CHF values are generated for given input values. The mean absolute percentage error (MAPE) and R^2 (coefficient of determination) are calculated as fitness parameters. A grid search was carried out to find the optimum number of layers, nodes, learning rate, and batch size that resulted in the best MAPE and R^2 values.

The MAPE is a metric used to evaluate model accuracy. It measures the average absolute percentage difference between the actual and predicted values. The formula for calculating MAPE is:

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^{n} \left| \frac{CHF_i - \widehat{CHF}_i}{CHF_i} \right| \times 100\%$$

where CHF_i represents the actual CHF values in a test set, $\widehat{CHF}_i$ represents the predicted values generated by cGAN, and n is the total number of samples in the test set. Lower MAPE values indicate higher accuracy, meaning that the predicted CHF values are closer to the actual values. R^2 is the popular statistical measure that represents the proportion of the variance in the dependent variable that is predictable from the independent variables in a regression model. It is often used as a measure of how well the regression model fits the data, with higher values indicating a better fit.

3.1. Data Assessment Cases

Table II illustrates three different cases investigated in this research work. In Case 1, we aim to assess the impact of training dataset size on cGAN performance for generating new synthetic data in the test (unseen) CHF values. First, we randomly take 20% of the total dataset as the test set (4,291 points). The training dataset size systematically varied, ranging from 80% of the total data (17,162 samples) to only 100 samples. The test set is fixed across every change in the training data size in Case 1. For example, whether cGAN is trained with 200 samples or 10,000 samples, it will be evaluated based on the same test set.

Table II. Description of the test cases used to evaluate GAN data needs

Case	Description
Case 1	Test dataset size is 20% of total data. Training dataset size varies from 100 samples to 17,162 samples (80% of total data).
Case 2	Scenario 1: Five selected experiment setups are used and combined (total 12,936 samples). The training dataset is 80% and the test dataset is the remaining 20%.
	Scenario 2: An experiment setup is taken as the training dataset, the remaining four experiment setups are used as a test dataset, and performance metrics are calculated by taking the average over the four test sets.
Case 3	Scenario 1: Four out of five input parameters are used, and one parameter was excluded. Five models can be trained where cGAN performance was measured by swapping the input parameter to be excluded.
	Scenario 2: Three input parameters are used, and two input parameters are excluded. Ten combinations are investigated to measure the effect of different pairs of excluded parameters.

Case 2 is dedicated to examining the possibility of generating CHF data of a specific experiment setup after training cGAN with a dataset sourced from another experiment setup. Five experimental setups are chosen for this case study out of 59 setups due to the larger number of CHF measurements available in those setups. These experiments are Alekseev et al., Becker et al., Kirillov et al., Smolin et al., and Zenkevich et al. [19]. Case 2 includes two scenarios as indicated in Table II. In Scenario 1, all data from the five experimental setups are combined, 20% is taken as a test set and the other 80% is used as training. In Scenario 2, one experiment (e.g., Alekseev et al.) was taken as the training dataset, and the other four experiments (e.g., Becker et al., Kirillov et al., Smolin et al., and Zenkevich et al.) were the test datasets. The overall cGAN performance was determined by taking the average of the four unseen experiments. The comparison of Scenario 1 and 2 results would indicate whether cGAN can generalize well to new experiments.

Finally, Case 3 explores the effect of the input parameters (conditions) on CHF generation quality. In the first scenario of Case 3, four input parameters were used, while one input parameter was left. For example, cGAN will use pressure, mass flux, inlet temperature, and diameter as inputs to generate CHF values, while skipping heated length (L), and so on for every other input parameter. Accordingly, five separate runs were conducted for five input parameters being held out in a one-at-a-time fashion. In the second scenario, three input parameters were used, and two input parameters were left. Ten different combinations were examined to assess the impact of various pairs of unused parameters on the cGAN model's performance. Case 3 aims to elucidate the influence of individual input parameters and their combinations on CHF generation accuracy using cGANs.

4. RESULTS AND DISCUSSIONS

For demonstration purposes, we select one case study from Table II to show in Figure 2 for a typical curve for the generator and discriminator losses during cGAN training. This figure was drawn from Case 1. The training loss, $\mathbb{E}_{z\sim p_z(z)}[\log(1-\mathcal{D}(\mathcal{G}(z|y)))]$, measures how good the generator in creating data that is close to real data. The discriminator loss, Eq.(1), quantifies how well the discriminator is doing in distinguishing real from fake data. As the training progresses, the discriminator loss should increase and the generator loss should decrease as given in Figure 2. Note that the scale of the discriminator loss deterioration is lower than the generator loss improvement, which could further normalize with longer training. Nevertheless, such loss curve still yields competitive GAN models based on the metrics that will be reported later in this section.

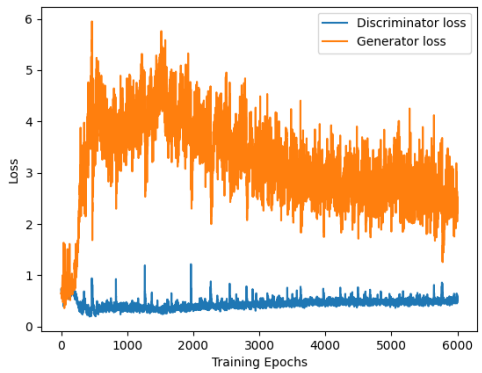

Figure 2. Generator and discriminator loss during training for a sample from Case 1 in Table II

Table III presents the performance metrics MAPE and R^2 values found using cGAN for different training dataset sizes. The number of layers, nodes, learning rate, and batch size that resulted in the best MAPE and R-squared values are also tabulated. For better comprehension, the comparison is also shown in Figure 3(a). The findings reveal that, as the training dataset size decreases, there was a gradual decline in the overall cGAN performance. MAPE increased nearly twofold from 9.08% with the maximum training dataset to 19.53% when the training dataset size was reduced to 100 samples. MAPE ranges from around 9–10% for larger dataset sizes (17,162, 10,000, 2,000, and 1,000), 13.68% for 500 points, 17.24% for 200 points, to approximately 20% for the smallest dataset size. This suggests that smaller dataset sizes result in less accurate predictions by GenAI. The metric values after using 1000 training samples started to saturate as the changes were marginal. The same conclusion can also be seen for R^2 values with a range between 0.828 for the smallest training sizes and 0.962 for the largest training sizes. With more data available for training, the model can learn more patterns and relationships in the data, leading to more accurate predictions and a better ability to generate more accurate synthetic CHF data.

Table III. Performance metrics for Case 1 on varying the training dataset size

Training Dataset Size	Learning Rate	Layers	Nodes	Batch Size	MAPE (%)	R^2
17,162	0.001	3	300	128	9.08	0.962
10,000	0.001	4	200	128	9.94	0.958
2,000	0.001	4	200	128	9.38	0.960
1,000	0.001	3	250	128	9.66	0.959
500	0.001	3	200	64	13.68	0.926
200	0.001	3	200	64	17.24	0.909
100	0.001	3	200	32	19.53	0.828

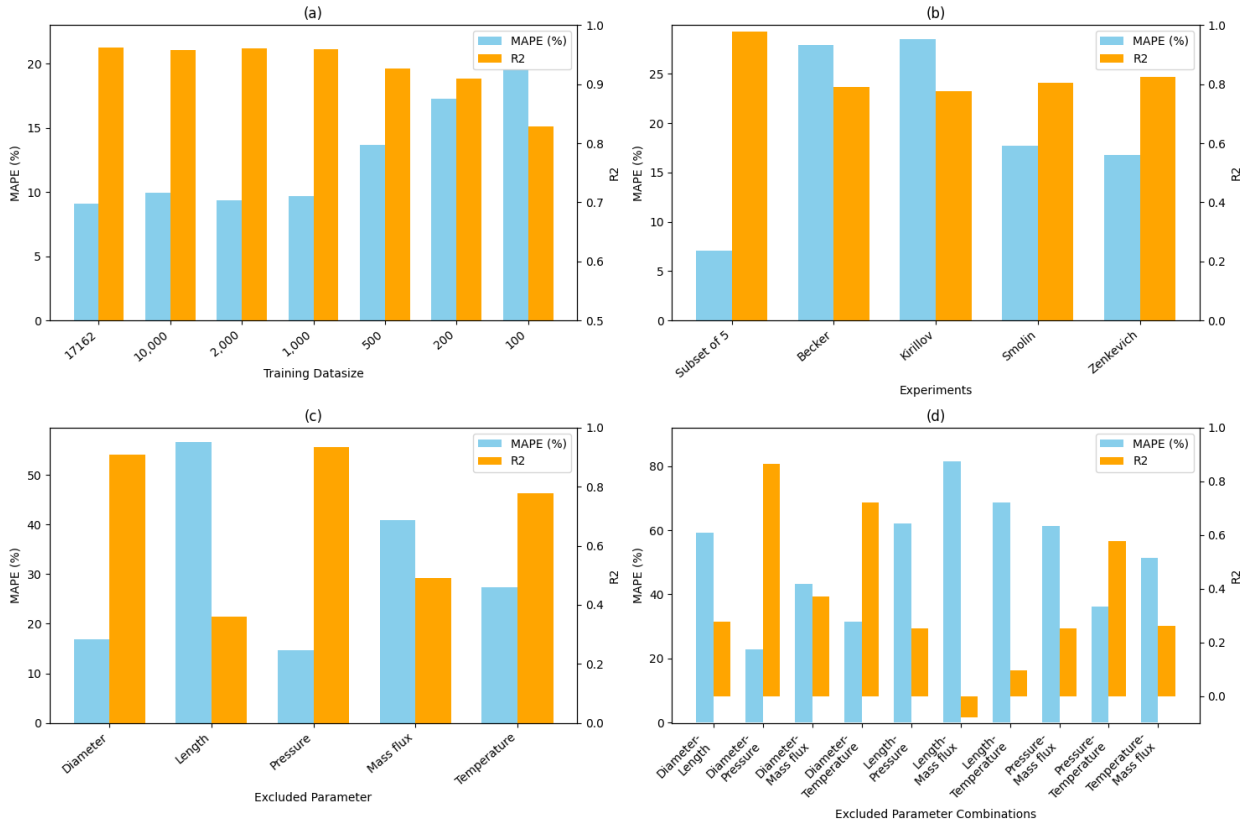

Figure 3. Comparison of MAPE and R^2 for (a) Case 1 on varying the training dataset size, (b) Case 2 on varying the experiments used in training and testing, (c) Case 3 Scenario 1 on excluding one input parameter at a time, and (d) Case 3 Scenario 2 on excluding two input parameters at a time.

The results in Table IV illustrate the performance metrics, including MAPE and R^2 values, achieved using a cGAN for different experiments based on Case 2 tests. To enhance clarity, the comparison is visually represented in Figure 3(b). For benchmarking purposes, Case 2 Scenario 1 is referred to as "Subset of 5" in Table IV and Figure 3(b), where all experiments are combined and 20% are used as a test set from all experiments. From the results, it can be seen that Scenario 1 led to a very good performance with MAPE $= 7.04\%$ and $R^2 = 0.977$.

Next, the fitness metrics were calculated for Case 2 Scenario 2 as in Table II, where one experiment setup was used as a training set and the other four were used as test sets. The dataset from Smolin et al. and Zenkevich et al. [19] have a relatively fair MAPE of 17.72% and 16.77% and R^2 of 0.805 and 0.824, respectively, which indicate a fair accuracy of generation when Smolin and Zenkevich measurements are used to generate the other four experiments. Two other experiment setups (i.e., Becker and Kirillov [19]) show lower accuracy when being used as training data. For example, Kirillov metrics are a MAPE of 28.51% and R^2 = 0.776. cGAN with Alekseev data completely failed to reproduce any of the four other experiments. This is because Alekseev's experiments do not adequately encompass the experimental conditions in the other setups, causing cGAN to heavily extrapolate in its generated data.

A noteworthy finding from Case 2 highlights a clear limitation of generative models, particularly cGAN

in our context, which often rely on exposure to a large number of samples from various configurations. Although Scenario 1 yielded commendable metrics when cGAN has seen a lot of data from all five experiments, cGAN's performance declined by at least 10% when tasked with replicating experiment setups it had not encountered previously. This highlights that we need to be very careful about how we use GAN or GenAI in synthetic data generation and how much margin we have in extrapolation.

Table IV. Performance metrics for Case 2 on varying the experiments used in training and testing

Experiment	MAPE (%)	R^2
Subset of 5	7.04	0.977
Alekseev	271.33	-5.109
Becker	27.91	0.789
Kirillov	28.51	0.776
Smolin	17.72	0.805
Zenkevich	16.77	0.824

Table V showcases the fitness metrics of cGAN for Case 3 Scenario 1 when four input parameters out of five were used during training. A visual depiction of the comparison is provided in Figure 3(c). The bar graph shows that a higher R^2 value was achieved when the diameter and pressure were not used as inputs, implying less impact on the cGAN generation ability of CHF. However, when the heated length or mass flux is excluded, cGAN performance significantly declined, showing poor CHF predictions. The inlet temperature has a moderate impact on CHF and comes in the middle of the five parameters.

Table V. Performance metrics for Case 3 Scenario 1 on excluding one input parameter at a time

Excluded Parameter	Learning Rate	Layers	Nodes	Batch Size	MAPE (%)	R^2
Diameter	0.001	3	200	128	16.84	0.910
Length	0.001	3	300	128	56.69	0.361
Pressure	0.001	3	200	128	14.68	0.935
Mass Flux	0.001	5	300	128	40.90	0.490
Temperature	0.001	4	250	128	27.33	0.779

To study the effect of parameter combinations on cGAN, the findings of Case 3 Scenario 2 are documented in Table VI and Figure 3(d). First, cGAN performance in all cases of Scenario 2 is much lower than in Scenario 1, implying that any feature exclusion could have a detrimental impact on cGAN generation accuracy, let alone multiple feature exclusions. Matching our conclusions from Case 3 Scenario 1, the combination of diameter and pressure did not cause a large reduction in performance metrics, while the length-mass flux pair caused a huge drop in R^2 (down to -0.08). This indicates that length and mass flux strongly influence the generation of accurate CHF values.

5. CONCLUSIONS

In this study, the authors examined how well GenAI techniques, particularly cGAN, perform in real-world engineering applications. We focused on the data requirements of cGAN when applied to CHF synethic data generation. We utilized a dataset comprising over 20,000 real experimental CHF measurements. Through three designed experiments, we explored the behavior of cGAN when varying the size of the training dataset, training cGAN on data from different experimental sources to generate new data from unseen experimental sources, and excluding various combinations of input parameters from the training to assess their impact

Table VI. Performance metrics for Case 3 Scenario 2 on excluding two input parameters at a time

Excluded Parameter Pair	Learning Rate	Layers	Nodes	Batch Size	MAPE (%)	R^2
Diameter-Length	0.001	5	250	128	59.31	0.276
Diameter-Pressure	0.001	3	350	128	22.75	0.864
Diameter-Mass Flux	0.001	4	350	128	43.30	0.370
Diameter-Temperature	0.001	5	200	128	31.48	0.721
Length-Pressure	0.001	5	200	128	62.18	0.254
Length-Mass Flux	0.001	3	300	128	81.56	-0.08
Length-Temperature	0.001	4	350	128	68.71	0.097
Pressure-Mass Flux	0.001	4	350	128	61.46	0.253
Pressure-Temperature	0.001	3	350	128	36.32	0.578
Temperature-Mass Flux	0.001	4	200	128	51.44	0.262

on cGAN's accuracy in data generation. The findings revealed that cGAN's performance decreases with smaller training dataset sizes, indicating its significant data dependency for reliable performance. When trained on data from certain experiments, cGAN performed well in predicting similar data but poorly when trained on different experiment sources. For example, when trained on data from Smolin et al. experiments, cGAN performed fairly well in predicting the data from Becker et al., Kirillov et al., and Alekseev et al. but provided very poor performance when trained with Alekseev et al. data and tasked to predict other experiment setups. In either case, cGAN performance was lower compared to when it was trained on samples from all experiments at the same time. In analyzing feature importance, we observed that cGAN relies heavily on input parameters such as mass flux and heated length for accurate CHF generation, while parameters like diameter and pressure have less influence. Inlet temperature showed a moderating effect.

For future research, we plan on extending the approach to probabilistic GANs by incorporating Bayesian techniques into the generator and discriminator. This adaptation aims to enable GANs to account for measured uncertainty in the data and generate CHF values within acceptable measurement uncertainty bounds provided in the CHF dataset.

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